# Evaluation runs: 2pc-le (election)

generated 2026-11-08 from brain-data 522afaee3 2026-11-08 12:20:57 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): leader-conf: expected conforms; ring-conf: expected conforms; leader-silent: expected violation; leader-split: expected violation; leader-split: expected conforms (one participant: nothing to split); ring-split: expected violation; ring-split: expected conforms (one participant: nothing to split). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair leader-conf, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 2 | 3 | brain03, 2026-11-03 |
| 3 | 18 | 588 | conforms | 0.1 | 4 | 1 h, 250 GB | 6 | 93 | brain03, 2026-11-03 |
| 4 | 96 | 1839744 | conforms | 22.9 | 22 | 1 h, 250 GB | 24 | 300 | brain03, 2026-11-03 |
| 5 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair ring-conf, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 0.0 | 4 | 1 h, 250 GB | 2 | 7 | brain03, 2026-11-03 |
| 3 | 18 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-03 |
| 4 | 96 | 192 | conforms | 0.1 | 4 | 1 h, 250 GB | 24 | 1452 | brain03, 2026-11-03 |
| 5 | 600 | 1920 | conforms | 0.8 | 6 | 1 h, 250 GB | 120 | 7260 | brain03, 2026-11-03 |
| 6 | 4320 | 23040 | conforms | 104.0 | 14 | 1 h, 250 GB | 720 | 1814760 | brain03, 2026-11-03 |
| 7 | 35280 | 322560 | conforms | 1760.5 | 30 | 1 h, 250 GB | 5285 | 66628084 | brain03, 2026-11-03 |
| 8 | 322560 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair leader-silent, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | 3 | 12 | brain03, 2026-11-03 |
| 3 | 18 | 196 | 196 | violation | 0.2 | 4 | 1 h, 250 GB | 196 | 3528 | brain03, 2026-11-03 |
| 4 | 96 | 459936 | 459936 | violation | 422.1 | 23 | 1 h, 250 GB | 459936 | 44153856 | brain03, 2026-11-03 |
| 5 | | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair leader-split, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 9 | violation | 0.0 | 4 | 1 h, 250 GB | 9 | 36 | brain03, 2026-11-03 |
| 3 | 18 | 882 | 882 | violation | 0.8 | 4 | 1 h, 250 GB | 882 | 15876 | brain03, 2026-11-03 |
| 4 | 96 | 2759616 | 2759616 | violation | 2213.8 | 28 | 1 h, 250 GB | 2759616 | 264923136 | brain03, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 6 | violation | 0.0 | 4 | 1 h, 250 GB | 6 | 24 | brain03, 2026-11-03 |
| 3 | 18 | 28 | 28 | violation | 0.1 | 4 | 1 h, 250 GB | 28 | 504 | brain03, 2026-11-03 |
| 4 | 96 | 120 | 120 | violation | 0.7 | 6 | 1 h, 250 GB | 120 | 11520 | brain03, 2026-11-03 |
| 5 | 600 | 496 | 496 | violation | 19.0 | 12 | 1 h, 250 GB | 496 | 297600 | brain03, 2026-11-03 |
| 6 | 4320 | 2016 | 2016 | violation | 543.1 | 17 | 1 h, 250 GB | 2016 | 8709120 | brain03, 2026-11-03 |
| 7 | 35280 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo, ring split 10 fifo.t16.p.r3, ring conf 10 fifo.t16.p.r3, ring conf 11 fifo, ring split 11 fifo, ring split 11 fifo.t16.p.r3.

skipped: a smaller size had reached its limit: leader silent 6 fifo, leader split 6 fifo, leader conf 6 fifo, leader split 7 fifo, leader silent 7 fifo, leader conf 7 fifo, ring split 8 fifo, leader split 8 fifo, leader silent 8 fifo, leader conf 8 fifo, leader split 9 fifo, ring split 9 fifo, leader silent 9 fifo, ring conf 9 fifo, leader conf 9 fifo, ring conf 10 fifo, ring split 10 fifo.

## stateful

### stateful: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 1.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 18 | 588 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 96 | 1839744 | conforms | 25.5 | 23 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 18 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 96 | 192 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 600 | 1920 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 6 | 4320 | 23040 | conforms | 4.4 | 6 | 1 h, 250 GB | brain02, 2026-11-03 |
| 7 | 35280 | 322560 | conforms | 33.8 | 38 | 1 h, 250 GB | brain02, 2026-11-03 |
| 8 | 322560 | 5160960 | conforms | 288.2 | 151 | 1 h, 250 GB | brain02, 2026-11-03 |
| 9 | 3265920 | 92897280 | conforms | 1893.5 | 1438 | 48 h, 250 GB | brain05, 2026-11-02 |
| 10 | 36288000 | 1857945600 | conforms | 45668.0 | 19992 | 48 h, 250 GB | brain05, 2026-11-02 |
| 12 | | | oom (250 GB) | | | 48 h, 250 GB | brain05, 2026-11-03 |
| 13 | | | oom (250 GB) | | | 48 h, 250 GB | brain01, 2026-11-07 |

### stateful: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 18 | 196 | 196 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 96 | 459936 | 459936 | violation | 9.2 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 9 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 18 | 882 | 882 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 96 | 2759616 | 2759616 | violation | 39.5 | 24 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 18 | 28 | 28 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 96 | 120 | 120 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 600 | 496 | 496 | violation | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 6 | 4320 | 2016 | 2016 | violation | 1.1 | 6 | 1 h, 250 GB | brain02, 2026-11-03 |
| 7 | 35280 | 8128 | 8128 | violation | 4.8 | 12 | 1 h, 250 GB | brain02, 2026-11-03 |
| 8 | 322560 | 32640 | 32640 | violation | 31.4 | 147 | 1 h, 250 GB | brain02, 2026-11-03 |
| 9 | 3265920 | 130816 | 130816 | violation | 151.7 | 1328 | 48 h, 250 GB | brain03, 2026-11-06 |
| 10 | 36288000 | 523776 | 523776 | violation | 1620.9 | 17884 | 48 h, 250 GB | brain03, 2026-11-06 |
| 11 | | | | oom (250 GB) | | | 48 h, 250 GB | brain04, 2026-11-07 |
| 13 | | | | oom (250 GB) | | | 48 h, 250 GB | brain05, 2026-11-03 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): ring split 10 fifo.t16.p.r3, ring conf 10 fifo.t16.p.r3, ring split 11 fifo, ring conf 12 fifo, ring split 12 fifo, ring conf 14 fifo, ring split 14 fifo, ring conf 15 fifo, ring split 15 fifo, ring split 16 fifo, ring conf 16 fifo, ring conf 17 fifo, ring split 17 fifo.

skipped: a smaller size had reached its limit: leader silent 6 fifo, leader split 6 fifo, leader conf 6 fifo, leader split 7 fifo, leader silent 7 fifo, leader conf 7 fifo, leader split 8 fifo, leader silent 8 fifo, leader conf 8 fifo, leader split 9 fifo, leader silent 9 fifo, leader conf 9 fifo.

left out by the table rules (ok): ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo.

left out by the table rules (timeout): ring conf 10 fifo, ring conf 11 fifo.

the tool ended with an error (exit-101): ring split 12 fifo.

## offline

### offline: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 18 | 588 | conforms | 0.2 | 8 | 4 h, 250 GB | 0.0 | 0.2 | 0.0 | merge | brain01, 2026-11-05 |
| 4 | 96 | 1839744 | conforms | 26.1 | 25 | 4 h, 250 GB | 0.0 | 26.1 | 0.0 | merge | brain01, 2026-11-05 |

### offline: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 18 | 24 | conforms | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 4 | 96 | 192 | conforms | 0.2 | 8 | 4 h, 250 GB | 0.0 | 0.2 | 0.0 | merge | brain01, 2026-11-05 |
| 5 | 600 | 1920 | conforms | 0.4 | 8 | 4 h, 250 GB | 0.1 | 0.3 | 0.1 | merge | brain01, 2026-11-05 |
| 6 | 4320 | 23040 | conforms | 3.9 | 12 | 4 h, 250 GB | 0.7 | 3.6 | 0.3 | merge | brain01, 2026-11-05 |
| 7 | 35280 | 322560 | conforms | 32.8 | 101 | 4 h, 250 GB | 3.9 | 30.0 | 2.8 | merge | brain01, 2026-11-05 |
| 8 | 322560 | 5160960 | conforms | 231.3 | 1378 | 4 h, 250 GB | 36.5 | 202.6 | 28.7 | merge | brain01, 2026-11-05 |
| 9 | 3265920 | 92897280 | conforms | 4557.6 | 13323 | 4 h, 250 GB | 337.5 | 4219.0 | 338.6 | merge | brain01, 2026-11-05 |

### offline: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 1 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 18 | 196 | 2 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 96 | 459936 | 6 | violation | 14.1 | 24 | 4 h, 250 GB | 0.0 | 14.1 | 0.0 | merge | brain01, 2026-11-05 |

### offline: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 3 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 18 | 882 | 9 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 4 | 96 | 2759616 | 36 | violation | 67.4 | 27 | 4 h, 250 GB | 0.0 | 67.4 | 0.0 | merge | brain01, 2026-11-05 |

### offline: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 3 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 18 | 28 | 7 | violation | 0.2 | 8 | 4 h, 250 GB | 0.0 | 0.2 | 0.0 | merge | brain01, 2026-11-05 |
| 4 | 96 | 120 | 15 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-05 |
| 5 | 600 | 496 | 31 | violation | 0.2 | 8 | 4 h, 250 GB | 0.1 | 0.2 | 0.0 | merge | brain01, 2026-11-05 |
| 6 | 4320 | 2016 | 63 | violation | 0.9 | 16 | 4 h, 250 GB | 0.7 | 0.6 | 0.2 | merge | brain01, 2026-11-05 |
| 7 | 35280 | 8128 | 127 | violation | 6.2 | 70 | 4 h, 250 GB | 3.9 | 2.0 | 2.4 | merge | brain01, 2026-11-05 |
| 8 | 322560 | 32640 | 255 | violation | 58.2 | 1165 | 4 h, 250 GB | 36.5 | 7.2 | 21.7 | merge | brain01, 2026-11-05 |
| 9 | 3265920 | 130816 | 511 | violation | 471.6 | 11668 | 48 h, 250 GB | 233.7 | 14.5 | 237.9 | merge | brain03, 2026-11-06 |
| 10 | 36288000 | 523776 | 1023 | violation | 6475.3 | 132550 | 48 h, 250 GB | 2913.4 | 35.7 | 3561.9 | merge | brain03, 2026-11-06 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo, ring split 10 fifo.vs-ring-conf.p.r3, ring conf 10 fifo.vs-ring-conf.p.r3, ring conf 11 fifo, ring split 11 fifo.vs-ring-conf.p.r3, ring split 12 fifo.

left out by the table rules (ok): ring conf 4 fifo, leader split 4 fifo, leader conf 4 fifo, leader silent 4 fifo, ring split 4 fifo, ring conf 5 fifo, ring split 5 fifo, ring split 6 fifo, ring conf 6 fifo, ring conf 7 fifo, ring split 7 fifo, ring split 8 fifo, ring conf 8 fifo, ring split 9 fifo.

skipped: a smaller size had reached its limit: leader silent 5 fifo, leader conf 5 fifo, leader split 5 fifo, leader silent 6 fifo, leader split 6 fifo, leader conf 6 fifo, leader split 7 fifo, leader silent 7 fifo, leader conf 7 fifo, leader split 8 fifo, leader silent 8 fifo, leader conf 8 fifo, leader split 9 fifo, leader silent 9 fifo, leader conf 9 fifo, ring conf 10 fifo.

left out by the table rules (memory): ring split 9 fifo, ring conf 9 fifo.

left out by the table rules (timeout): ring split 10 fifo.

refused (an input missing at run time): ring split 11 fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 3.2 | 160 | 10 min, 250 GB | 0.1 | 3.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 18 | 588 | conforms | 5.0 | 266 | 10 min, 16 GB | 0.1 | 5.0 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 3.3 | 165 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 18 | 24 | conforms | 3.6 | 176 | 10 min, 16 GB | 0.1 | 3.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 96 | 192 | conforms | 4.7 | 255 | 10 min, 16 GB | 0.1 | 4.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5 | 600 | 1920 | conforms | 12.8 | 2709 | 48 h, 80 GB | 0.9 | 12.8 | 0.0 | merge | pinky28, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 1 | violation | 2.9 | 174 | 10 min, 250 GB | 0.1 | 2.9 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 18 | 196 | 2 | violation | 4.0 | 166 | 10 min, 16 GB | 0.1 | 4.0 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 3 | violation | 3.1 | 170 | 10 min, 16 GB | 0.1 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 18 | 882 | 9 | violation | 5.4 | 287 | 10 min, 16 GB | 0.1 | 5.4 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 3 | violation | 3.1 | 170 | 10 min, 16 GB | 0.1 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 18 | 28 | 7 | violation | 3.5 | 166 | 10 min, 16 GB | 0.1 | 3.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 96 | 120 | 15 | violation | 4.0 | 170 | 10 min, 16 GB | 0.1 | 4.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5 | 600 | 496 | 31 | violation | 7.8 | 202 | 48 h, 80 GB | 0.9 | 7.8 | 0.0 | merge | pinky28, 2026-11-07 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): ring conf 4 fifo, leader split 4 fifo, leader conf 4 fifo, leader silent 4 fifo, ring split 4 fifo, ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo, ring conf 11 fifo, ring split 11 fifo.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 3.1 | 159 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 18 | 588 | conforms | 3.4 | 174 | 10 min, 16 GB | 3.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 3.1 | 159 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 18 | 24 | conforms | 3.4 | 174 | 10 min, 16 GB | 3.4 | 0.0 | 0.0 | in-memory | brain01, 2026-11-01 |
| 4 | 96 | 192 | conforms | 3.8 | 160 | 10 min, 16 GB | 3.8 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 1 | violation | 3.1 | 159 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 18 | 196 | 2 | violation | 3.4 | 174 | 10 min, 16 GB | 3.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 3 | violation | 3.1 | 159 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 18 | 882 | 9 | violation | 3.4 | 174 | 10 min, 16 GB | 3.4 | 0.2 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 3 | violation | 3.1 | 159 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 18 | 28 | 7 | violation | 3.4 | 174 | 10 min, 16 GB | 3.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 4 | 96 | 120 | 15 | violation | 3.8 | 160 | 10 min, 16 GB | 3.8 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): ring conf 4 fifo, leader split 4 fifo, leader conf 4 fifo, leader silent 4 fifo, ring split 4 fifo, ring conf 5 fifo, ring split 5 fifo, ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo.

skipped: a smaller size had reached its limit: ring split 11 fifo, ring conf 11 fifo.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 0.7 | 174 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 18 | 588 | conforms | 3.0 | 256 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 0.7 | 152 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 18 | 24 | conforms | 0.9 | 150 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 4 | 96 | 192 | conforms | 2.0 | 279 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 3 | violation | 0.7 | 148 | 1 h, 250 GB | 44 | brain05, 2026-11-02 |
| 3 | 18 | 196 | 196 | violation | 1.5 | 184 | 1 h, 250 GB | 56 | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 9 | violation | 0.7 | 153 | 1 h, 250 GB | 48 | brain05, 2026-11-02 |
| 3 | 18 | 882 | 882 | violation | 3.9 | 290 | 1 h, 250 GB | 66 | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 6 | violation | 0.8 | 152 | 1 h, 250 GB | 62 | brain05, 2026-11-02 |
| 3 | 18 | 28 | 28 | violation | 0.8 | 150 | 1 h, 250 GB | 54 | brain05, 2026-11-02 |
| 4 | 96 | 120 | 120 | violation | 1.4 | 207 | 1 h, 250 GB | 67 | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): leader conf 2 fifo, ring conf 2 fifo, ring split 2 fifo, leader silent 2 fifo, leader split 2 fifo, ring split 3 fifo, leader conf 3 fifo, leader split 3 fifo, ring conf 3 fifo, leader silent 3 fifo, ring conf 4 fifo, ring split 4 fifo, ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo, ring conf 11 fifo, ring split 11 fifo.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | conforms | 3.1 | 155 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 18 | 588 | conforms | 3.4 | 170 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **ring-conf**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 4 | conforms | 3.1 | 155 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 18 | 24 | conforms | 3.4 | 170 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 4 | 96 | 192 | conforms | 3.8 | 156 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 3 | violation | 3.1 | 155 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 18 | 196 | 196 | violation | 3.4 | 170 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **leader-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 9 | 9 | violation | 3.1 | 155 | 1 h, 250 GB | 6 | brain04, 2026-11-01 |
| 3 | 18 | 882 | 882 | violation | 3.5 | 170 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **ring-split**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 6 | 6 | violation | 3.1 | 155 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 18 | 28 | 28 | violation | 3.4 | 170 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 4 | 96 | 120 | 120 | violation | 3.8 | 156 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** left out by the table rules (ok): leader conf 2 fifo, ring conf 2 fifo, ring split 2 fifo, leader silent 2 fifo, leader split 2 fifo, ring split 3 fifo, leader conf 3 fifo, leader split 3 fifo, ring conf 3 fifo, leader silent 3 fifo, ring conf 4 fifo, ring split 4 fifo.

planned but never scheduled (grid points beyond the ladder): ring split 9 fifo, ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo, ring conf 11 fifo, ring split 11 fifo.

# context-free twins online, Rust implementation (TraceForge) against the Java port's export

## context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ring-split**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 9 | 130816 | 130816 | conforms | 435.9 | 13750 | 48 h, 80 GB | | pinky07, 2026-11-05 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** planned but never scheduled (grid points beyond the ladder): ring conf 9 fifo, ring split 10 fifo, ring conf 10 fifo.